

Synopsys[®] Cognitive Services

PGVector Docker Deployment Guide

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Contents

PGVector Docker Deployment Guide	2
• 1. Prerequisites	2
• 1.1 Hardware Requirements	2
• 1.2 System Requirements Verification	2
• 1.3 Access Requirements	2
• 1.4 Obtain the Synopsys Installer & SPF	3
• 2. Deployment	4
• 2.1 Installed Package Layout	4
• 2.2 Navigate to Working Directory	5
• 2.3 Load Container Image	5
• 2.4 (Optional) Replace Image Reference	6
• 2.5 Generate TLS Certificates	6
• 2.6 Deploy PgVector	7
• 2.7 Validation	8
• 2.7.1 Install psql Client	8
• 2.7.2 Connect	9
• 3. Cleanup	9

PGVector Docker Deployment Guide

This guide explains how to install and run the SCS PgVector (PostgreSQL + vector extension) service using Docker Compose in an on-premises environment.

1. Prerequisites

Before beginning the installation, ensure you have the following resources and environment readiness.

1.1 Hardware Requirements

Service	CPU	Memory	Disk (Data Dependent)	GPU	GPU Memory
PgVector	4	Varies	Varies	Not required	Not required

Note: Size for your workload. Disk growth depends on table count, index strategy, and vector dimensions.

1.2 System Requirements Verification

- Docker Engine: v24.0.7 or later
See: <https://docs.docker.com/engine/install/ubuntu/>
- Docker Compose: v2.24.0 or later
See: <https://docs.docker.com/compose/install/linux/>
- Minimum 4 CPU cores (more for concurrency)
- Minimum 16 GB RAM recommended
- User with permission to run Docker / manage volumes

1.3 Access Requirements

- Synopsys Installer binary (current version, for example `SynopsysInstaller_v5.9.run`)

- PgVector SPF file: `scs_onprem_docker-compose_pgvector.spf`
- Local directory with read/write access for installation & data

1.4 Obtain the Synopsys Installer & SPF

1.4.1 Download

1. Obtain `SynopsysInstaller_v<version>.run` via Synopsys support portal or representative.
2. Download the PgVector SPF: `scs_onprem_docker-compose_pgvector.spf`.
3. Place all SPF files in a single directory (example: `~/downloads/<release>/`).

1.4.2 Install the Synopsys Installer

This step prepares the installer for use on your system.

1. Make the installer executable:

```
chmod +x SynopsysInstaller_v5.9.run
```

2. Run the installer:

```
./SynopsysInstaller_v5.9.run
```

3. When prompted, choose an install directory (example: `/home/user/installer`).

Example output (truncated):

```
Installing Synopsys Installer 5.9 into the directory  
'/home/user/installer'...  
Installation complete.
```

1.4.3 Install PgVector via Installer

This step installs the PgVector package using the Synopsys Installer.

1. Navigate to your home or working directory:

```
cd ~
```

2. Run the installer:

```
<installer_path>/installer/installer
```

3. Respond to prompts:

- Source directory: path containing the SPF files
- Target directory: where to install PgVector
- Confirm installation summary (type `yes`)

Sample summary:

```
##### Installation Summary #####
Product:      scs_pgvector
Release:      <version>
Platforms:    linux64 common
Disk Space:   34 MB
Target Dir:
/home/user/scs_pgvector/scs_onprem_docker_compose_pgvector/<version>
#####
Accept, Install? [yes]: yes
```

2. Deployment

2.1 Installed Package Layout

After installation, the directory structure resembles:

```
scs-onprem-docker-pgvector-package
├── <version>
│   ├── admin
│   │   ├── license
│   │   └── site_info
│   ├── install.log
│   ├── linux64.placeholder
│   └── scs_onprem_docker_compose_pgvector
│       ├── config
│       │   ├── pg_hba.conf
│       │   └── postgresql.conf
│       ├── docker-compose.yaml
│       ├── docs
│       │   └── SCS-Deployment-Guide-Docker-PgVector.pdf
│       ├── pgvector-pg17.tar
│       └── scripts
│           └── postgres-init.sh
```

2.2 Navigate to Working Directory

- Change to the installation directory:

```
cd
<installation_root>/scs_onprem_docker_compose_pgvector/<version>
```

2.3 Load Container Image

- Load the PgVector image (PostgreSQL 17) from the provided archive:

```
docker load -i pgvector-pg17.tar
```

Air-gapped: Ensure all dependency images referenced in `docker-compose.yaml` are pre-loaded locally.

Upgrade Notice: PgVector / PostgreSQL upgraded from 16 → 17. Backup existing data before deploying the new major version and perform a proper `pg_upgrade` if

migrating.

2.4 (Optional) Replace Image Reference

If the compose file references a registry image and you are offline, edit `docker-compose.yml` to point to the locally loaded image tag shown by `docker load`.

2.5 Generate TLS Certificates

This step creates the required certificates for secure connections.

1. Create a directory for certificates and enter it:

```
mkdir certs && cd certs
```

2. Set certificate details:

```
export SSL_COUNTRY="US" SSL_STATE="California" SSL_CITY="San  
Francisco" \  
    SSL_ORGANIZATION="My Company" SSL_UNIT="IT Department"  
SSL_DOMAIN="myserver.local" filename=server
```

3. Generate CA key:

```
openssl genrsa -out ca.key 4096
```

4. Generate CA certificate:

```
openssl req -key ca.key -new -x509 -days 365 -sha256 -extensions  
v3_ca -out ca.pem -subj \  
  
"/C=$SSL_COUNTRY/ST=$SSL_STATE/L=$SSL_CITY/O=$SSL_ORGANIZATION/OU  
=$SSL_UNIT/CN=$SSL_DOMAIN"
```

5. Generate server key and CSR:


```
openssl req -out $filename.csr -new -newkey rsa:2048 -nodes -  
keyout $filename.key -subj \  
  
"/C=$SSL_COUNTRY/ST=$SSL_STATE/L=$SSL_CITY/O=$SSL_ORGANIZATION/OU  
=$SSL_UNIT/CN=$SSL_DOMAIN"
```

6. Sign the server certificate:

```
openssl x509 -req -days 360 -in $filename.csr -CA ca.pem -CAkey  
ca.key -CAcreateserial -out $filename.crt -sha256
```

7. Return to the working directory:

```
cd ..
```

Expected files: `certs/ca.pem`, `certs/server.crt`, `certs/server.key`.

2.6 Deploy PgVector

This step starts the PgVector service using Docker Compose.

- Set a secure password and start services:

```
PGVECTOR_PORT=5432 PGVECTOR_ROOT_PASSWORD=<password> docker  
compose -f docker-compose.yaml up -d
```

2.6.1 Verify Port Availability (Optional)

- Check if the required port is available:

```
netstat -tln | grep 5432
```

No output means the port is free.

2.6.2 Monitor Startup

1. List running containers:

```
docker ps
```

2. View logs for the PgVector container:

```
docker logs -f <pgvector_container_name>
```

Look for:

```
LOG: database system is ready to accept connections
```

2.7 Validation

2.7.1 Install psql Client

Choose the appropriate method for your OS:

1. Ubuntu / Debian:

```
sudo apt-get update  
sudo apt-get install postgresql-client-17
```

If you get "package not found" or wrong version:

```
curl -fsSL https://www.postgresql.org/media/keys/ACCC4CF8.asc |  
sudo gpg --dearmor -o  
/etc/apt/trusted.gpg.d/apt.postgresql.org.gpg  
echo "deb http://apt.postgresql.org/pub/repos/apt $(lsb_release -  
cs)-pgdg main" | sudo tee /etc/apt/sources.list.d/pgdg.list  
sudo apt-get update  
sudo apt-get install postgresql-client-17
```

2. macOS (Homebrew):

```
brew install libpq
```

Add to PATH if needed:

```
echo 'export PATH="/usr/local/opt/libpq/bin:$PATH"' >> ~/.bashrc
```

3. **Windows:** Install [pgAdmin](#) (GUI) or the full PostgreSQL installer (includes `psql`) from [PostgreSQL](#).

4. **Air-gapped:** Download source (for example, <https://www.postgresql.org/ftp/source/v17.5/>) then:

```
make -C src/bin install
make -C src/include install
make -C src/interfaces install
make -C doc install
```

2.7.2 Connect

- Connect to the database:

```
psql -h <hostname> -p 5432 -U postgres
```

Enter the password. Sample banner:

```
psql (17.x)
SSL connection (protocol: TLSv1.3, cipher:
TLS_AES_256_GCM_SHA384)
postgres=#
```

3. Cleanup

1. Stop services (retains volumes):

```
docker compose -f docker-compose.yaml down
```

2. Remove containers and volumes (permanent data loss):

```
docker compose -f docker-compose.yaml down -v
```

Warning: `-v` deletes persisted database data.